

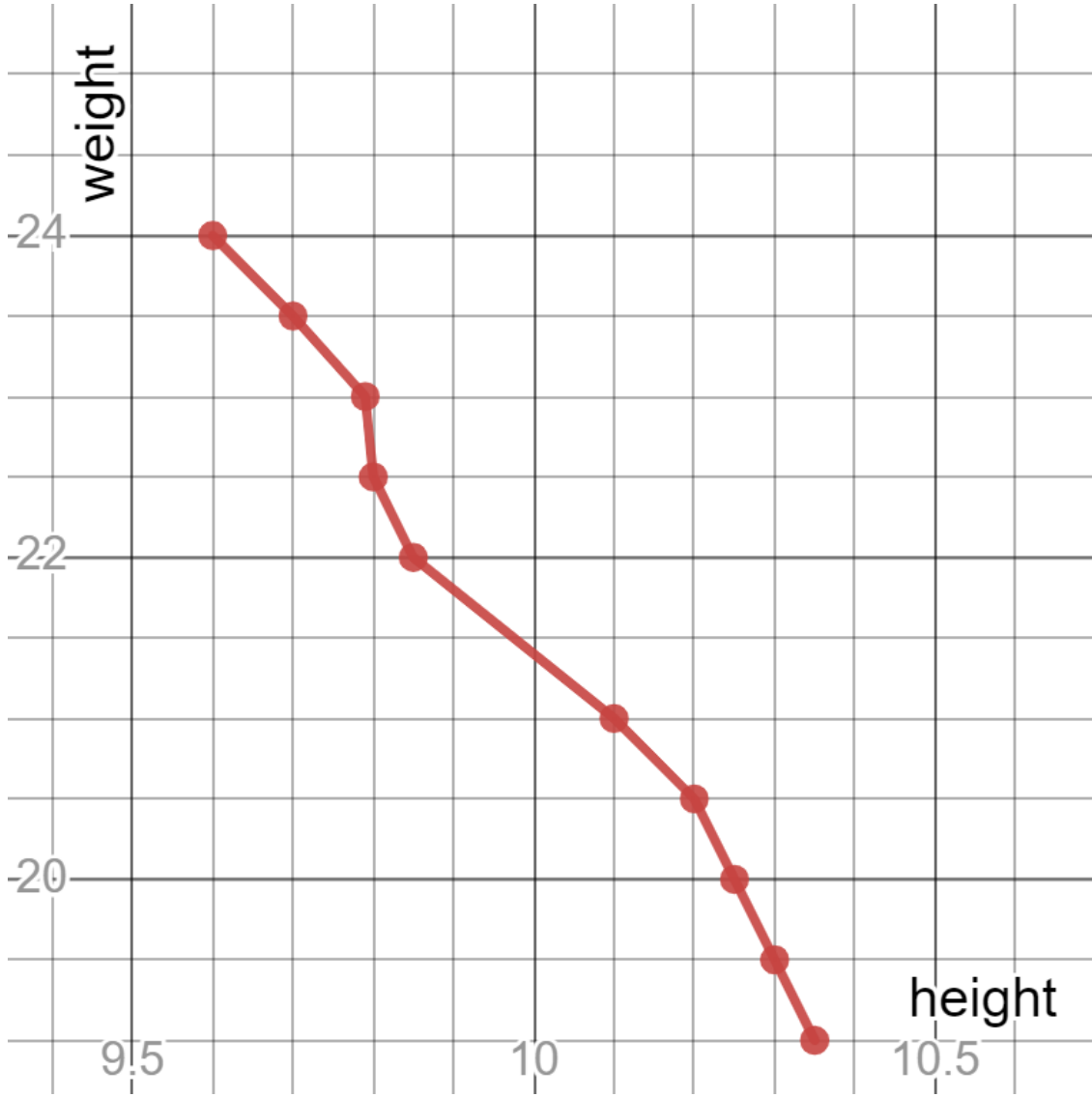
Algebra 1 Unit 12 Assessment Guide

General Information	Throughout middle school, students were introduced to different types of data displays. In Algebra 1, the benchmarks MA.912.DP.1.1, MA.912.DP.1.2, and MA.912.DP.1.3 incorporate students' prior knowledge to select an appropriate method to represent the data, interpret data distributions and their components and quantities, and to explain correlation. Although not exclusively stated in MA.912.DP.1.1, clarification 3 indicates that students may be expected to create an appropriate display. For students who struggle creating displays, teachers may wish to support these students using lessons from the middle school courses. Teachers should also note that the benchmarks MA.912.DP.1.1, MA.912.DP.1.2, MA.912.DP.1.3, and MA.912.DP.1.4 are also in Probability and Statistics Honors. This means that an Algebra 1 student would not need to achieve the same depth of understanding of these benchmarks as a student who is taking Probability and Statistics Honors. At times, this means the depth to which to teach can be hard to gauge. The benchmark MA.912.DP.1.4 is especially difficult to gauge because of the statistics that is within the benchmark. Teachers may wish to utilize the Algebra 1 B1G-M to further gauge the expectation of an Algebra 1 student. What is not clearly stated in the B1G-M for MA.912.DP.1.3 but is found in the draft test item specifications is that in order to show causation, assessment items must explicitly state whether a controlled study or experiment was performed. Notice that this was done on this assessment in question 5. Often students learn the phrase, "correlation does not mean causation" whereas if correlation is shown within a controlled study or experiment, then it is possible to show causation. The length of the assessment is 7 pages long because the benchmarks include displays and require the student to explain. To best model how Florida's end-of-course assessment has students justify their answers, this assessment has the student complete statements. These also require a lot of space. Teachers may determine that it is best to divide the assessment into parts or to use the questions as formative assessments.
Calculator Information	The questions on the unit assessment are calculator.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	1a	Benchmark(s)	MA.912.DP.1.1
	☐ Box plot ☒ Bar graph ☐ Histogram ☐ Line graph ☐ Scatter plot ☒ Circle graph ☐ Stem-and-leaf ☒ Frequency table ☐ Line plot (Dot plot) ☒ Relative Frequency table	Recommended Scoring (9 Total Points)	Consider giving students 3 points for each correct answer. Consider a deduction of 3 points for each incorrect selection. Students who select all of the options should not receive any points.
Question	1b	Benchmark(s)	MA.912.DP.1.1
	Because the data set of the water districts is ☐ numerical ☒ categorical and ☐ bivariate, ☒ univariate, the displays chosen in part a can be used.	Recommended Scoring (3 Total Points)	Partial credit is not recommended.
Question	1c	Benchmark(s)	MA.912.DP.1.1
	☒ Box plot ☐ Bar graph ☒ Histogram ☐ Line graph ☐ Scatter plot ☐ Circle graph ☒ Stem-and-leaf ☒ Frequency table ☒ Line plot (Dot plot) ☐ Relative Frequency table	Recommended Scoring (15 Total Points)	Consider giving students 3 points for each correct answer. Consider a deduction of 3 points for each incorrect selection. Students who select all of the options should not receive any points.

Question	1d	Benchmark(s)	MA.912.DP.1.1
	Because the data set of the water districts is ☐ numerical ☐ categorical and ☐ bivariate, ☐ univariate, the displays chosen in part c can be used.	Recommended Scoring (3 Total Points)	Partial credit is not recommended.
Question	2a	Benchmark(s)	MA.912.DP.1.4
	1651 to 1799 students	Recommended Scoring (6 Total Points)	Consider giving students 3 points for each endpoint of the range.
Question	2b	Benchmark(s)	MA.912.DP.1.4
	39,381 to 42,923 students	Recommended Scoring (6 Total Points)	Consider giving students 3 points for each endpoint of the range.

Question	3	Benchmark(s)	MA.912.DP.1.2
Complete the statements. a. The measure of center that best summarizes the cost of a healthy diet is the ☐ mean ☒ median because display ☐ X ☒ Y ☐ Z shows that the ☒ mean ☐ median would be influenced by ☒ outliers. ☐ the range. ☐ the standard deviation. ☐ the interquartile range. b. Displays ☐ X and Y ☐ X and Z ☐ Y and Z ☒ X, Y, and Z can be used to show that the shape of the data is ☒ skewed ☐ bell-shaped which indicates that the ☐ range ☐ outliers ☐ standard deviation ☒ interquartile range would be the best measure of variability.		Recommended Scoring (6 Total Points)	Consider giving students 3 points for each correctly completed statement. Partial credit for each statement is not recommended because the student should get all parts correct for the statement to make sense.

Question	4	Benchmark(s)	MA.912.DP.1.2																					
 The line graph displays the relationship between height and weight. The x-axis, labeled 'height', has major tick marks at 9.5, 10, and 10.5. The y-axis, labeled 'weight', has major tick marks at 20, 22, and 24. The graph consists of 10 data points connected by a red line. The points show a general upward trend, with a slight dip in the middle. <table><tr><th>Height</th><th>Weight</th></tr><tr><td>9.6</td><td>24.0</td></tr><tr><td>9.7</td><td>23.5</td></tr><tr><td>9.8</td><td>23.0</td></tr><tr><td>9.8</td><td>22.5</td></tr><tr><td>9.8</td><td>22.0</td></tr><tr><td>10.1</td><td>21.0</td></tr><tr><td>10.2</td><td>20.5</td></tr><tr><td>10.3</td><td>20.0</td></tr><tr><td>10.4</td><td>19.5</td></tr><tr><td>10.5</td><td>19.0</td></tr></table>	Height	Weight	9.6	24.0	9.7	23.5	9.8	23.0	9.8	22.5	9.8	22.0	10.1	21.0	10.2	20.5	10.3	20.0	10.4	19.5	10.5	19.0	Recommended Scoring (34 Total Points)	Consider giving students 2 points for each point that is correctly plotted, 4 points for each correct label, and 6 points for connecting consecutive points to create the line graph.
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10.4	19.5																							
10.5	19.0																							

Question	5a	Benchmark(s)	MA.912.DP.1.3
	a. Complete the statement. The value of the correlation coefficient shows that ☐ cell phone use ☐ poor sleep quality ☐ the age of a student ☒ addition to smartphones is ☒ correlated ☐ not correlated with ☐ cell phone use. ☒ poor sleep quality. ☐ the age of a student. ☐ addiction to smartphones.	Recommended Scoring (4 Total Points)	Partial credit is not recommended.
Question	5b	Benchmark(s)	MA.912.DP.1.3
	b. Complete the statement. The researchers ☒ can ☐ cannot claim that ☐ cell phone use ☐ poor sleep quality ☐ the age of a student ☒ addition to smartphones causes ☐ cell phone use ☒ poor sleep quality ☐ the age of a student ☐ addiction to smartphones because ☒ the study was a controlled study. ☐ correlation does not imply causation.	Recommended Scoring (4 Total Points)	Partial credit is not recommended.

Question	6a	Benchmark(s)	MA.912.DP.1.2																								
	882	Recommended Scoring (4 Total Points)	Partial credit is not recommended.																								
Question	6b	Benchmark(s)	MA.912.DP.1.2																								
<table><thead><tr><th>Category</th><th>Number of Students</th><th>Category</th><th>Number of Students</th></tr></thead><tbody><tr><td>Individual sports</td><td>637</td><td>Team sports</td><td>392</td></tr><tr><td>Art</td><td>196</td><td>Music</td><td>490</td></tr><tr><td>Dance</td><td>196</td><td>Computer programing</td><td>49</td></tr><tr><td>Video games</td><td>245</td><td>Debate team</td><td>147</td></tr><tr><td>Chess club</td><td>98</td><td></td><td></td></tr></tbody></table> b. Complete the statement. Based on original survey results the circle graph would have ☒ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 sectors that are labeled as 10% or less and ☒ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 labeled as more than 10%.		Category	Number of Students	Category	Number of Students	Individual sports	637	Team sports	392	Art	196	Music	490	Dance	196	Computer programing	49	Video games	245	Debate team	147	Chess club	98			Recommended Scoring (6 Total Points)	Consider giving students 3 points for each correct answer box.
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